KAIZEN MATHS

Lesson starter

Roots of Equations and Transformations of Roots

Exam-Style Roots

Name

S1

Use repeated roots to form the equation.

A cubic equation has roots 3, 3 and -5. Form the equation.

S2

Use the difference between roots to find the parameter.

The equation $x^2 + kx + 44 = 0$ has two positive roots that differ by 7. Find k.

S3

Use conjugate roots to form real polynomial equations.

A cubic equation with real coefficients has roots $2 + 3i$, $2 - 3i$ and -2. Form the equation.

S4

Use root relationships to evaluate advanced symmetric expressions.

The roots of $x^3 - 2x^2 - 11x + 12 = 0$ are α, β and γ. Find $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$.

Show your working

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KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Exam-Style Roots

Name _____

EA1

Use repeated roots to form the equation.

A cubic equation has roots 2, 2 and -1. Form the equation.

EA2

Use the difference between roots to find the parameter.

The equation $x^2 + kx + 55 = 0$ has two positive roots that differ by 6. Find k.

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KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Exam-Style Roots

Name _____

EB1

Use conjugate roots to form real polynomial equations.

A cubic equation with real coefficients has roots -1 + 2i, -1 − 2i and 2. Form the equation.

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KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Exam-Style Roots

Name _____

EA1

Use repeated roots to form the equation.

A cubic equation has roots 2, 2 and -1. Form the equation.

EA2

Use the difference between roots to find the parameter.

The equation $x^2 + kx + 55 = 0$ has two positive roots that differ by 6. Find k.

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KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Exam-Style Roots

Name _____

EB1

Use conjugate roots to form real polynomial equations.

A cubic equation with real coefficients has roots -1 + 2i, -1 − 2i and 2. Form the equation.

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Teacher answers & working

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S1

Use repeated roots to form the equation.

A cubic equation has roots 3, 3 and -5. Form the equation.

Answer:

$x^3 - x^2 - 21x + 45 = 0$

Working:

The root 3 is repeated, so the roots are α = 3, β = 3, γ = -5.

$\alpha + \beta + \gamma = 3 + 3 + -5 = 1.$

$\alpha\beta + \alpha\gamma + \beta\gamma = (3)(3) + (3)(-5) + (3)(-5) = -21.$

$\alpha\beta\gamma = (3)(3)(-5) = -45.$

Therefore $x^3 - x^2 - 21x + 45 = 0$.

S2

Use the difference between roots to find the parameter.

The equation $x^2 + kx + 44 = 0$ has two positive roots that differ by 7. Find k.

Answer:

$k = -15$

Working:

Let the roots be α and β, with α > β.

$\alpha\beta = 44$ and $\alpha - \beta = 7.$

$(\alpha + \beta)^2 = (\alpha - \beta)^2 + 4\alpha\beta.$

$(\alpha + \beta)^2 = 7^2 + 4(44) = 225.$

Since both roots are positive, α + β = 15.

For $x^2 + kx + 44 = 0$, $\alpha + \beta = -k.$

So −k = 15, giving k = -15.

Teacher answers & working

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S3

Use conjugate roots to form real polynomial equations.

A cubic equation with real coefficients has roots 2 + 3i, 2 − 3i and -2. Form the equation.

Answer:

$x^3 - 2x^2 + 5x + 26 = 0$

Working:

Complex roots of a polynomial with real coefficients occur in conjugate pairs.

The conjugate pair gives $x^2 - (4)x + 13$.

The remaining root -2 gives the factor $(x + 2)$.

Multiply: $(x^2 - 4x + 13)(x + 2)$.

Therefore $x^3 - 2x^2 + 5x + 26 = 0$.

S4

Use root relationships to evaluate advanced symmetric expressions.

The roots of $x^3 - 2x^2 - 11x + 12 = 0$ are α, β and γ. Find $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$.

Answer:

169

Working:

Notice that $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$.

From the equation, $\alpha + \beta + \gamma = 2$.

$\alpha\beta + \alpha\gamma + \beta\gamma = -11$.

$\alpha\beta\gamma = -12$.

So the expression is $(-11)^2 - 2(-12)(2) = 169$.

EA1

Use repeated roots to form the equation.

A cubic equation has roots 2, 2 and -1. Form the equation.

Answer:

$x^3 - 3x^2 + 4 = 0$

Working:

The root 2 is repeated, so the roots are $\alpha = 2$, $\beta = 2$, $\gamma = -1$.

$\alpha + \beta + \gamma = 2 + 2 + -1 = 3$.

$\alpha\beta + \alpha\gamma + \beta\gamma = (2)(2) + (2)(-1) + (2)(-1) = 0$.

$\alpha\beta\gamma = (2)(2)(-1) = -4$.

Therefore $x^3 - 3x^2 + 4 = 0$.

Teacher answers & working

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EA2

Use the difference between roots to find the parameter.

The equation $x^2 + kx + 55 = 0$ has two positive roots that differ by 6. Find k.

Answer:

$k = -16$

Working:

Let the roots be α and β, with α > β.

$\alpha\beta = 55$ and $\alpha - \beta = 6$.

$(\alpha + \beta)^2 = (\alpha - \beta)^2 + 4\alpha\beta$.

$(\alpha + \beta)^2 = 6^2 + 4(55) = 256$.

Since both roots are positive, α + β = 16.

For $x^2 + kx + 55 = 0$, $\alpha + \beta = -k$.

So −k = 16, giving k = -16.

EB1

Use conjugate roots to form real polynomial equations.

A cubic equation with real coefficients has roots -1 + 2i, -1 − 2i and 2. Form the equation.

Answer:

$x^3 + x - 10 = 0$

Working:

Complex roots of a polynomial with real coefficients occur in conjugate pairs.

The conjugate pair gives $x^2 - (-2)x + 5$.

The remaining root 2 gives the factor (x − 2).

Multiply: $(x^2 + 2x + 5)(x - 2)$.

Therefore $x^3 + x - 10 = 0$.